

Jonathan Lozano

PhD Candidate in Theoretical Physics

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Education

- 2023–present **Ph.D. in Physics**, *The University of Texas at Austin, USA*
Advisor: Dr. Antonios Alvertis.
GPA: 3.9/4.00. Research in condensed matter and many-particle physics.
- 2016–2021 **B.Sc. in Physics**, *National Autonomous University of Mexico (UNAM), Mexico City*,
GPA: 9.39/10.0 — *High Honors*
Advisor: Dr. Manuel Torres Labansat. Coursework included two graduate-level courses in quantum field theory and a graduate course in differential geometry and topology for physicists.
Thesis: Spontaneous symmetry breaking and extended field configurations in a scalar theory subject to a potential with two families of vacuum states.

Publications

- 2026 **J. Lozano-Mayo**. Generalized virial identities: radial constraints for solitons, instantons, and bounces. *J. High Energ. Phys.* **04** (2026) 058. doi:10.1007/JHEP04(2026)058
- 2026 **J. Lozano-Mayo** and M. Torres-Labansat. Multi-kinks and composite oscillons in a commensurable and non degenerate double sine-Gordon model. *J. High Energ. Phys.* **09** (2026) 218. doi:10.1007/JHEP09(2026)218
- 2025 J. J. Ziegler, K. Freese, **J. Lozano**, G. Montefalcone. Explaining the “too massive” high-redshift galaxies in JWST data: a numerical study of three effects and a simple relation. (*In revision*).
- 2021 **J. Lozano-Mayo** and M. Torres-Labansat. Kink solutions in a generalized scalar ϕ_G^4 field model. *J. Phys. Commun.* **5** (2021) 025004. doi:10.1088/2399-6528/abdd83
- 2024-2025 *ALICE Collaboration*: Contributed to ITS performance analysis, and hadron- D^0 azimuthal-correlation studies as service work during December 2024-October 2025. Publications bearing my name under ALICE’s collective authorship policy are not listed here.

Conference Presentations

- 2025 **J. L. Mayo**. $h-D^0$ angular correlations and ΔROF performance studies at ITS2. ALICE-USA Meeting.
- 2021 **J. L. Mayo**. *Multi-solitones en teorías escalares de campo con vacíos no-degenerados: el modelo doble de sine-Gordon*. LXVI National Physics Congress, Mexican Physical Society.

Research Experience

- 2026 **Strongly interacting electron-phonon systems**, *UT Austin*, Advisor: Prof. Antonios Alvertis
Studied the properties of strongly interacting electron-phonon systems using tensor network methods.

- 2025 **h - D^0 Angular Correlations**, *ALICE Collaboration*, Advisor: Prof. Christina Markert
Studied the properties of charm production across distinct color environments using ALICE Run-3 Pb-Pb collisions, characterizing the azimuthal angular difference between a high-momentum trigger hadron and a lower-momentum associated D^0 meson.
- 2024 **JWST Early Massive Galaxies and Top-Heavy IMFs**, *UT Austin*, Advisor: Prof. Katherine Freese
Modeled the spectral energy distributions of high-redshift galaxies with the population-synthesis code PÉGASE to constrain the parameter space in which JWST observations remain consistent with the standard Λ CDM cosmology.
- 2021–2022 **Multi-kinks in Scalar Field Theories with Non-degenerate Vacua**, *UNAM*, Advisor: Dr. Manuel Torres Labansat
Investigated the formation of static n -kink structures in models with deformed potentials. Performed numerical simulations in Julia to characterize the binding forces between kinks and derived an analytical relation for the energy of the static multi-kink.
- 2020–2021 **Quantum Renormalization of Kink Masses and Virial Relations**, *UNAM*, Advisor: Dr. Manuel Torres Labansat
Renormalized the one-loop quantum mass correction of kink configurations arising from the topology of a generalized ϕ^4 potential using perturbative techniques. Established the existence of static multi-kink configurations via asymptotic analysis and developed a framework for their stability.
- 2018–2019 **Photon Wave Function**, *UNAM*, Advisor: Dr. Manuel Torres Labansat
Examined whether a well-defined photon wave function can be constructed from the photon dispersion relation. Used group-theoretic methods to build the quantum operators and Lorentz transformations acting on a proposed six-component wave function, then derived the Lagrangian density and verified the expected symmetries of the theory.
- 2019 **Time-of-Flight Mass Spectrometry**, *UNAM*, Advisor: Dr. Juan López Patiño
Produced proton-air collisions with a low-energy linear collider and analyzed the reaction products using the time-of-flight mass spectrometry technique.

Teaching & Academic Service

- 2026 **Teaching Assistant — Graduate Quantum Mechanics**, *UT Austin*
Mentored students and graded assignments for a graduate course on Quantum Mechanics.
- 2025 **Lecturer — Quantum Field Theory I**, *ICTP (International Centre for Theoretical Physics)*
Delivered an introductory quantum field theory course as part of the International Centre for Theoretical Physics outreach initiative.
- 2024 **Teaching Assistant — Advanced Particle Physics**, *ICTP*
Led problem-set discussions and graded assignments for a graduate course on advanced topics in quantum field theory.
- 2025 **Teaching Assistant — Waves and Optics**, *UT Austin*
Held weekly discussion sections and graded coursework spanning wave phenomena, optics, and electromagnetism.
- 2024–2025 **Teaching Assistant — Modern Physics**, *UT Austin*
Led weekly discussion sections and graded coursework spanning statistical mechanics, quantum mechanics, and special relativity.
- 2023 **Teaching Assistant — Electromagnetism Laboratory (PHY 105N)**, *UT Austin*
Led two weekly laboratory sections and supported the experimental setups.
- 2022 **Teaching Assistant — Thermodynamics**, *UNAM*
Held problem-solving sessions and created and graded homework and exams.

- 2020 **Teaching Assistant — Nuclear and Subnuclear Physics, UNAM**
Mentored students and graded homework and exams for a course taught by Prof. Manuel Torres Labansat.
- 2022 **Referee, *J. Phys. G: Nuclear and Particle Physics***
Reviewed submitted manuscripts.
- 2017 **Volunteer Instructor, UNAM School of Humanities and Sciences**
Taught physics and mathematics in an undergraduate preparation course.

Awards & Distinctions

- 2022 **DPG/IAPS PLANCKS Munich Travel Grant** — awarded to attend the PLANCKS competition at LMU Munich.
- 2022 **Winner, PLANCKS 2022 Mexican Preliminary** — placed 1st of 30 teams in the Mexican Physics Tournament (*Torneo Mexicano de Física*).
- 2021 **Juan Manuel Lozano Mejía Diploma** — awarded by the UNAM Institute of Physics to students with the most outstanding thesis research.
- 2021 **Excellence Cluster Fellowship** — PRISMA, Johannes Gutenberg University Mainz.
- 2021 **Top 10, PLANCKS 2021 (Porto)** — placed 7th of 50 teams in the international theoretical physics competition.
- 2021 **Winner, PLANCKS 2021 Mexican Preliminary** — selected for the first Mexican team to attend PLANCKS after placing 1st nationally.
- 2021 **Honorific Mention, B.Sc. thesis defense (UNAM)** — highest distinction awarded at defense.
- 2019 **Research Assistantship, UNAM Institute of Physics.**
- 2018 **AMC Scholarship** — Mexican Academy of Sciences, XXVIII Scientific Research Summer.
- 2018 **ICF-UNAM Scholarship** — VII Experimental Physics School.
- 2017 **IF-UNAM Scholarship** — XXV Physics School.

Additional Training

- 2025 Summer School on CMB Data Analysis — UT Austin.
- 2022 Summer School on Modeling and Tools for Data Analysis in Science — UANL.
- 2021 Bad Honnef Physics School (DPG) — Methods of Effective Field Theory and Lattice Field Theory.
- 2021 VII Mexican School on String Theory and Supersymmetry (MSSS) — University of Guanajuato.
- 2021 Mexican Astro-Particle School (MAPS) — University of Guanajuato.

Research Interests

Condensed matter physics · Many-particle physics · Quantum phenomena ·
Tensor networks for strongly correlated systems · Topological and non-topological
solitons · Heavy-flavor physics

Technical Skills

Programming Julia, Python, C++, Mathematica, $\text{\LaTeX}$, HTML

Numerical methods Tensor networks (DMRG), exact diagonalization, finite-difference PDE solvers

Scientific tools ROOT/O² (ALICE framework), QtiPlot, Origin, Tracker, ImageJ

Languages

Spanish (native) · **English** (C1) · **German** (A2) · **French** (A2)